

WASP-52b

R-0534

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_240825 · created 2026-07-21T18:41:53+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460547.825 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.029
Transit depth (Rp/Rs)^2	4.6 +/- 1.23 [%]
Orbital Inclination (inc)	85.48 +/- 1.75
Airmass coefficient 1 (a1)	1.513 +/- 0.017
Airmass coefficient 2 (a2)	-0.335 +/- 0.091
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	None
Optimal Aperture	4.48
Optimal Annulus	17.28
Transit Duration (day)	0.0641 +/- 0.0083

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.214 ± 0.029 vs 0.1646 (deviation 1.7σ)
Duration fit vs published	0.0641 d vs 0.0758 d (deviation 1.41σ)
Mid-time O–C	-17.1 min
Depth SNR	7.4
Residual scatter	1.12 %

Light curve

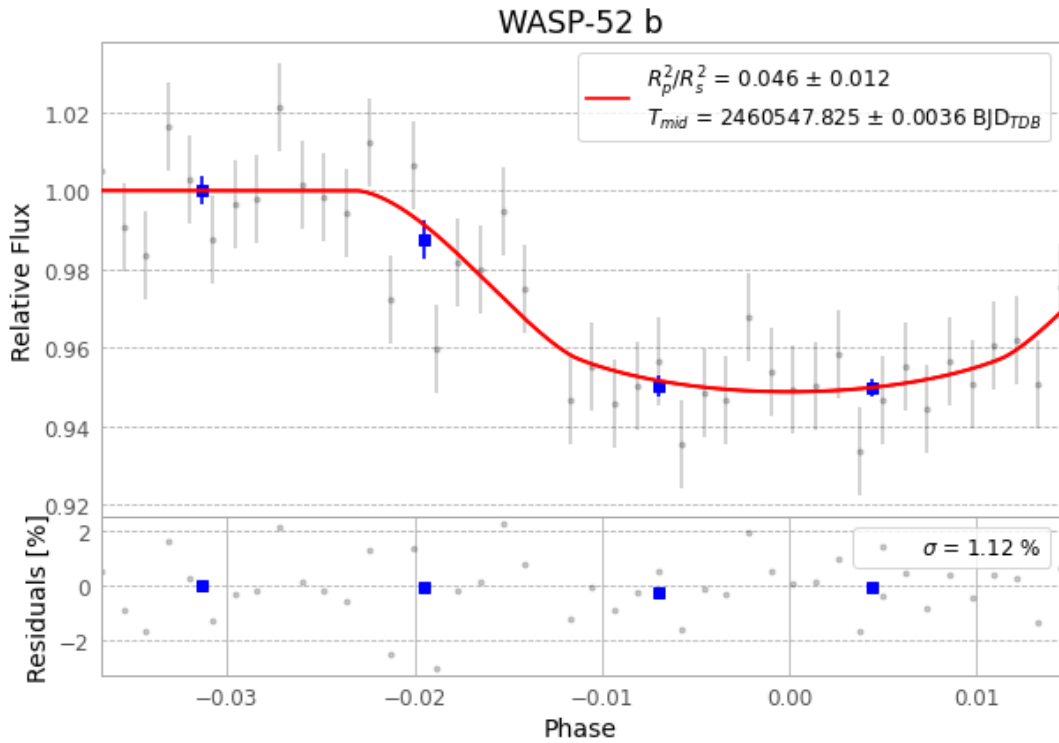


Figure 1 — FinalLightCurve_WASP-52 b_2024-08-24.png (SHA-256 3dc1f5f83a445032...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 abb62ea32115c717...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

46 FITS frames (30 MB), WASP-52240825060615.FITS → WASP-52240825082115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-240825.log (4bb10b7d4a85...), FinalLightCurve_WASP-52 b_2024-08-24.png (3dc1f5f83a44...), FinalLightCurve_WASP-52 b_2024-08-24.csv (1f47c847e656...)

Compute resources

Wall time 195.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 18:41Z.